The Three-Prime Running LCM: Complete Reasoning Record

Erdős Problem #269

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Abstract

For three pairwise distinct primes, the reciprocal running-LCM kernel has nonsingular minors of every order, with the same indices valid in every third-coordinate layer. Removing row and column factors leaves a binary floor carry. Its threshold columns give both the arbitrary-order rank obstruction and the exact rank of a finite sample. The normalised carry also has a sharp uniform finite-rank approximation barrier, although the original summable kernel admits finite-rank approximation in the summation norm. These are statements about representations, not irrationality of the resulting scalar sum.

For two distinct primes, both the repeated and de-duplicated running-LCM series are transcendental. They are nonconstant polynomials in one Hecke–Mahler value, whose transcendence follows from the theorem of Loxton and van der Poorten in the form given by Bugeaud and Laurent. Steve Fan posted the factorisation, reduction and conclusion first, on 26 June 2026. The argument recorded here uses that external theorem; it is not a Lean formalisation of the two-prime result.

For the repeated $\{2, 3, 5\}$ series, the complete record constructs the literal tail recurrence and proves that rationality forces eventually positive integral tails under a quadratic bound. Escape of the exact window residues is equivalent to irrationality. Finite certificates test that criterion and exclude bounded denominator ranges, but do not supply its unbounded quantifiers. The record gives the proofs, exact finite evidence and failed representation methods behind the short paper; the repeated-series problem with three or more primes remains open.

How the short paper uses this record. The short paper centres on the binary carry and its rank obstruction. Section 4 gives the rank and approximation arguments; Section 3 gives the two-prime comparison. Sections 5–7 recover the literal tails, denominator clearing and window criterion, while Section 8 retains the finite certificates. These are two distinct mathematical tasks: the rank argument excludes a separated representation, and the tail argument identifies the arithmetic condition still needed for irrationality. Neither substitutes for the other.

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

1 The problem, and what is settled

Let P be a finite set of primes with $|P| \geq 2$, and let $a_1 < a_2 < \cdots$ enumerate the positive integers all of whose prime factors lie in P. Erdős Problem #269 asks whether

$$\sum_{n \geq 1} \frac{1}{[a_1, \ldots, a_n]}$$

is irrational, where $[a_1, \ldots, a_n]$ is the least common multiple [1, p. 65][2, p. 106]. Bloom's catalogue records the problem as open [4]. This note settles every instance with $|P| = 2$, at the level of transcendence, and the unresolved finite cases begin at $|P| = 3$.

The restriction $|P| \geq 2$ is necessary. For $P = \{p\}$ the enumeration is $a_n = p^{n-1}$, so $[a_1, \ldots, a_n] = p^{n-1}$ and the sum is $p/(p-1)$. For infinite P the sum is always irrational, which Erdős calls a simple exercise [2, p. 106].

Write $\mathcal{R}_P$ for the sum above and $\mathcal{D}_P$ for the de-duplicated sum, in which each distinct value of the running least common multiple contributes its reciprocal once. The two differ because the running value is constant along stretches of the enumeration. In a letter written on 1 January 1973 Erdős recorded that he could prove irrationality of the de-duplicated sum [3, p. 335]. He states that assertion for a general finite list of primes and supplies no proof, so the historical record already covers $\mathcal{D}_P$ for every finite P; the open question addressed by the catalogue is $\mathcal{R}_P$.

Theorem 1.1 (two-prime transcendence). *Let p and q be distinct primes. Then $\mathcal{R}_{\{p,q\}}$ and $\mathcal{D}_{\{p,q\}}$ are transcendental.*

Section 3 proves this. Both values are explicit polynomials in one Hecke–Mahler boundary sum A, the coefficients are nonzero rationals, and transcendence of A is the theorem of Loxton and van der Poorten [11, p. 40, Theorem 8] in the modern form of Bugeaud and Laurent's Theorem 1.1 [10, p. 61, Theorem 1.1]. The evidence class is an ordinary proof in this paper over a cited external theorem, and no Lean declaration formalises it.

Attribution. Steve Fan posted the running-LCM identity for every $|P|$, the two-channel factorisation, the Hecke–Mahler reduction and the transcendence conclusion in the discussion thread of the problem's page on 26 June 2026 [12], with follow-up remarks extending the argument to arbitrary coprime pairs. The priority for the two-prime case is his. The proof below is independent and is given in full because it fixes the $|P| = 2$ boundary of the problem and because the same boundary sum drives the three-prime discussion. The identity of Theorem 2.1 is cited to him.

What the third prime does. Section 4 isolates the obstruction exactly. After separating row and column factors, the two-prime kernel is an outer product and the three-prime kernel retains one binary floor carry. That carry is enough to produce nonsingular minors of every order, uniformly in the third coordinate, and to make the normalised carry matrix inapproximable by finite separated sums below half its own jump. Sections 5 to 7 take a second route on the literal $\{2, 3, 5\}$ series: dyadic shells give an integer-coefficient tail recurrence, a strict endpoint clears the smooth part of a hypothetical denominator, and the surviving carry is trapped between a quadratic bound and a residue condition. Section 8 reports the finite evidence and its ceiling, and Section 9 states what remains.

Throughout, p, q, r are pairwise distinct primes. Call n *smooth* when $n = p^i q^j r^k$ for some $i, j, k \geq 0$; this is the smooth lattice value. For $x \geq 1$ write

$$\mathrm{L}(x) = \operatorname{lcm}\{\, n \leq x : \ n \text{ smooth} \,\}, \qquad \mathrm{H}(x) = p^{\lfloor \log_p x \rfloor} q^{\lfloor \log_q x \rfloor} r^{\lfloor \log_r x \rfloor},$$

the running least common multiple and the pure-power height, where $\lfloor \log_b x \rfloor$ is the largest e with $b^e \leq x$. Since $a_1, \ldots, a_n$ are exactly the smooth numbers up to a_n, we have $[a_1, \ldots, a_n] = \mathrm{L}(a_n)$. The reciprocal of the height at a smooth point is the lattice kernel

$$\mathrm{K}(i, j, k) = \frac{1}{\mathrm{H}(p^i q^j r^k)}.$$

Here "smooth" always means supported on the fixed prime set, and it is not the varying-bound notion counted by $\Psi(x, y)$ in the Dickman–Hildebrand theory [7, Theorem 1]; no smooth-number density asymptotic is used below. For $b \in \{p, q, r\}$ the set of positive powers of b is the *b-channel*.

The statement of Problem #269 has been formalised before, as a conjecture with an unfilled proof, in the *Formal Conjectures* collection [5]. That is a formal statement of the question up to a rational normalisation, since its Nat-indexed series includes the empty-prefix term, and its rational, irrational and infinite-prime assertions all end in `sorry`. Kovač and Tao [9] treat several irrationality problems of Erdős for series of unit fractions by elementary means; nothing from that work is used here.

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2 The finite geometry of the running value

The smooth numbers up to x are indexed by the exponent triples (i, j, k) with $i \leq \lfloor \log_p x \rfloor$, $j \leq \lfloor \log_q x \rfloor$, $k \leq \lfloor \log_r x \rfloor$ and $p^i q^j r^k \leq x$, the smooth prefix index set. Both conditions matter: the coordinate box is strictly larger than the prefix, since a product of three large pure powers can exceed x while each factor does not.

Theorem 2.1 (the running least common multiple)**.** *Let p, q, r be pairwise distinct primes and $x \geq 1$. Then* $\mathrm{L}(x) = \mathrm{H}(x)$.

Proof. Every smooth $n \leq x$ has exponents bounded by the corresponding integer logarithms, so $n \mid \mathrm{H}(x)$ and hence $\mathrm{L}(x) \mid \mathrm{H}(x)$. Conversely the three pure powers $p^{\lfloor \log_p x \rfloor}$, $q^{\lfloor \log_q x \rfloor}$ and $r^{\lfloor \log_r x \rfloor}$ are themselves smooth numbers not exceeding x, so each divides $\mathrm{L}(x)$, and distinct primes have coprime powers, so their product divides $\mathrm{L}(x)$ as well. □

Formalised as the running-lcm identity, from the pointwise divisibility, the divisibility into the height and the three pure-power memberships, namely the first, second and third. The unrestricted analogue is classical: iterating the prime-exponent maximum rule gives $\operatorname{lcm}(1,\ldots,N) = \prod_{t\le N} t^{\lfloor \log_t N\rfloor}$ over the primes $t \le N$ [6, Ex. 1.21(a), p. 22], equivalently $\log \operatorname{lcm}(1,\ldots,N) = \psi(N)$ [6, §4.2, p. 75], and Montgomery and Vaughan state the identity directly in Exercise 6.2.7 [8, p. 183]. Fan states the P-restricted form for every $|P|$ [12]. The pairwise distinctness hypothesis is used exactly once, in the coprimality step.

At $(p,q,r) = (2,3,5)$ the first ten values are

x	1	2	3	4	5	6	7	8	9	10
$\mathrm{L}(x)$	1	2	6	12	60	60	60	120	360	360

So $\mathrm{L}(6) = 4\cdot 3\cdot 5 = 60$, which exceeds 6: the running value at a smooth cutoff already contains powers of the other two primes that the cutoff itself does not. That small fact is the mechanism behind Section 4. Also $\mathrm{H}(x) \le x^3$, since each of the three factors is a power of its base not exceeding x; this is the cubic majorant, and the exponent is the number of generating primes.

Say that x and y lie in the same *logarithmic cell* when $\lfloor \log_b x\rfloor = \lfloor \log_b y\rfloor$ for each of $b = p,q,r$, the cell relation. By Theorem 2.1 the running value depends on x only through the three integer logarithms, so it is constant on cells and moves only where one logarithm moves.

Proposition 2.2 (constancy and jump ratios). *If $x,y \ge 1$ lie in the same logarithmic cell then $\mathrm{L}(x) = \mathrm{L}(y)$, and the same holds for the kernel at two smooth points of one cell. If $\lfloor \log_p y\rfloor = \lfloor \log_p x\rfloor + 1$ while the other two logarithms agree, then $\mathrm{L}(y) = p\,\mathrm{L}(x)$, and similarly with q or r in place of p.*

Proof. Both parts are immediate from Theorem 2.1: the height depends on x only through the three integer logarithms, and advancing one of them multiplies exactly one factor by its base. □

The first part is the cell constancy of the running value, over the cell constancy of the height and the cell constancy of the kernel. The jump statement is the first, second and third coordinate steps, over the corresponding height steps, which need no primality: the first, second and third.

Proposition 2.3 (jump count). *Let $n \ge 0$. The set of the first n positive powers of p, of q and of r has exactly $3n$ elements, and adjoining the common origin 1 gives exactly $3n+1$.*

Proof. Within one channel the powers $b, b^2, \ldots, b^n$ are distinct because $b \ge 2$. Across two channels a common value would be a positive power of two distinct primes, which unique factorisation forbids. Finally 1 is not a positive power of any prime. □

Formalised as the positive jump count and the jump count with the origin, over the channel cardinality, the channel disjointness and the exclusion of the origin; the channels are the positive power sets, whose union is the positive jump set and, with the origin adjoined, the full jump set. Two channels never meet, so at each positive pure power exactly one logarithm advances and the running value is multiplied there by the prime of that channel. Reading those multipliers in increasing order of the pure powers gives the *jump word* of L. At $\{2,3,5\}$ the pure powers in increasing order are $2,3,4,5,8,9,16,25,27,32,\ldots$, so the jump word begins

$$2,\ 3,2,\ 5,2,\ 3,2,\ 5,3,2,\ \ldots$$

grouped so that each group ends at a power of two, which is the grouping used in Section 5.

Two finite statements are recorded here because they are the exact shape of the counting used later. Write $\mathcal{B}(h_p,h_q,h_r)$ for the box of exponent triples with $i \le h_p$, $j \le h_q$, $k \le h_r$, the exponent box, and $F(H)$ for the set of its points whose height equals H, the height fibre.

Proposition 2.4 (finite normal form)**.** *For every box $\mathcal{B}$, $\sum_{(i,j,k)\in\mathcal{B}} \mathrm{K}(i,j,k) = \sum_H \#F(H)/H$, the outer sum ranging over the heights attained on $\mathcal{B}$.*

Proof. Partition $\mathcal{B}$ into the fibres of the height map. On $F(H)$ every summand is $1/H$, so the fibre contributes $\#F(H)/H$. □

Formalised as the height-fibre normal form, over the fibre sum and the point height. At $(2,3,5)$ on $\mathcal{B}(1,1,1)$ the eight smooth values $1,2,3,5,6,10,15,30$ carry heights $1,2,6,60,60,360,360,10800$, so two coefficients equal 2 and the identity reads $18421/10800$ on both sides. The identity is between two finite sums over a box, so it is not a statement about L at a cutoff. The same module records the one-step map $\tau(b,d,s) = b(s-d)$, the variable-base tail step, with the rewriting that names its expanded form; the orbits analysed in Section 5 are the actual ones, and no orbit of this abstract map is analysed.

Now fix an interval $[\lambda,\eta)$ and write $\mathcal{S}$ for the exponent triples of $\mathcal{B}(h_p,h_q,h_r)$ whose smooth value lies in it, the smooth exponent shell.

Lemma 2.5 (uniqueness in a short interval)**.** *Let $b \ge 1$ and $\eta \le b\lambda$. If $b^a w$ and $b^{a'} w$ both lie in $[\lambda,\eta)$ then $a = a'$.*

Proof. If $a < a'$ then $\eta \le b\lambda \le b^{a+1} w \le b^{a'} w < \eta$, which is impossible; the case $a > a'$ is symmetric. □

Proposition 2.6 (projection and the quadratic shell bound)**.** *If $\eta \le r\lambda$ then $\#\mathcal{S} \le (h_p+1)(h_q+1)$, and if $\eta \le p\lambda$ then $\#\mathcal{S} \le (h_q+1)(h_r+1)$. If moreover $\eta \le r\lambda$ and $h_p \le h_q \le h_r$ with $h_p + h_q + h_r = j$, then $9\,\#\mathcal{S} \le (j+3)^2$.*

Proof. Suppose $\eta \le r\lambda$. If two triples of $\mathcal{S}$ agree in their first two coordinates, Lemma 2.5 with $b = r$ and $w = p^i q^j$ forces their third coordinates to agree, so the projection forgetting the third coordinate is injective on $\mathcal{S}$ and its image lies in a rectangle with $(h_p+1)(h_q+1)$ points. The other case is the same with the first coordinate projected away. Under the sorting hypothesis the two surviving coordinates are the two smallest, so it suffices that $a \le b \le c$ with $a+b+c = j$ gives $9(a+1)(b+1) \le (j+3)^2$. From $a \le b \le c$ we get $a + 2b \le j$, so it is enough that $9(a+1)(b+1) \le (a+2b+3)^2$; writing $b = a + d$ with $d \ge 0$, the difference of the two sides is $d(3a+4d+3) \ge 0$. □

Formalised as the third-coordinate projection, the first-coordinate projection, the quadratic shell bound and the sorted quadratic estimate, over the short-interval uniqueness. The constant 9 is the square of the number of generating primes, and the last inequality of the proof is an equality when $h_p = h_q = h_r$, so no larger constant survives that step. The estimate uses no analytic input on the distribution of smooth numbers.

3 One Hecke–Mahler value controls both two-prime sums

Temporarily let $P = \{p, q\}$ with $p < q$, and write $L_{p,q}(t) = p^{\lfloor \log_p t \rfloor} q^{\lfloor \log_q t \rfloor}$, which is the running least common multiple of the $\{p, q\}$-smooth numbers up to t by the argument of Theorem 2.1 with one coordinate omitted. The de-duplicated sum retains the initial value 1 and one reciprocal for every later distinct running value, so

$$\mathcal{D}_{\{p,q\}} = 1 + \sum_{t \in \{p,p^2,\ldots\} \cup \{q,q^2,\ldots\}} \frac{1}{L_{p,q}(t)}, \qquad \mathcal{R}_{\{p,q\}} = \sum_{i,j \geq 0} \frac{1}{L_{p,q}(p^i q^j)}.$$

Proof of Theorem 1.1. Set

$$\theta = \frac{\log p}{\log q}, \qquad x = \frac{1}{p}, \qquad y = \frac{1}{q}, \qquad m_n = \lfloor n\theta \rfloor, \qquad \delta_n = m_{n+1} - m_n.$$

Here $0 < \theta < 1$, and θ is irrational, since a rational value would give $p^b = q^a$ for positive integers a, b. Consequently $\delta_n \in \{0, 1\}$. Put

$$A = \sum_{n \geq 0} x^n y^{m_n}, \qquad B_* = \sum_{n \geq 0} \delta_n x^n y^{m_n + 1}.$$

The initial value and the p-channel contribute A, since $L_{p,q}(p^n) = p^n q^{m_n}$. A power of q lies strictly between p^n and p^{n+1} exactly when $\delta_n = 1$, it is then q^{m_n+1}, and its post-jump reciprocal is $x^n y^{m_n+1}$; so the q-channel contributes B_* and $\mathcal{D}_{\{p,q\}} = A + B_*$. All these series converge absolutely.

Since $y^{m_{n+1}} - y^{m_n} = \delta_n y^{m_n}(y-1)$, an index shift gives $A - 1 - xA = x(y-1)B_*/y$, so $B_* = ((p-1)A - p)/(1-q)$ and

$$\mathcal{D}_{\{p,q\}} = \frac{(q-p)A + p}{q-1}. \tag{1}$$

At a smooth point, $\log_p(p^i q^j) = i + j/\theta$ and $\log_q(p^i q^j) = j + i\theta$, so $L_{p,q}(p^i q^j) = p^{i + \lfloor j/\theta \rfloor} q^{j + m_i}$ and absolute convergence permits the factorisation

$$\mathcal{R}_{\{p,q\}} = AC, \qquad C = \sum_{j \geq 0} y^j x^{\lfloor j/\theta \rfloor}.$$

For $j \geq 1$ put $n = \lfloor j/\theta \rfloor$. Then $n > j/\theta - 1$ gives $n\theta > j - \theta > j - 1$, and $n + 1 > j/\theta$ gives $(n+1)\theta > j$, while $n\theta \leq j$ and $(n+1)\theta < j + \theta < j + 1$; hence $m_n = j - 1$, $m_{n+1} = j$ and $\delta_n = 1$. Conversely every n with $\delta_n = 1$ arises from the unique $j = m_{n+1}$. Therefore $C = 1 + B_*$ and

$$\mathcal{R}_{\{p,q\}} = \frac{(p+q-1)A - (p-1)A^2}{q-1}. \tag{2}$$

It remains to prove that A is transcendental. For the Hecke–Mahler series

$$F_\theta(x,y) = \sum_{n\geq 1}\sum_{k=1}^{\lfloor n\theta\rfloor} x^n y^k$$

a finite geometric sum gives $((1-y)/y)F_\theta(x,y) = x/(1-x) - (A-1)$, that is

$$A = \frac{1}{1-x} - \frac{1-y}{y}F_\theta(x,y). \tag{3}$$

Bugeaud and Laurent's Theorem 1.1 states, in particular, that $F_\theta(\beta,\alpha)$ is transcendental when $\theta \in (0,1)$ is irrational, α and β are nonzero algebraic numbers, $|\beta| < 1$ and $|\beta\alpha^\theta| < 1$ [10, p. 61, Theorem 1.1]; the $\rho = 0$ case used here goes back to Loxton and van der Poorten [11, p. 40, Theorem 8]. Take $(\beta,\alpha) = (x,y)$: then $|\beta| = 1/p < 1$ and

$$|xy^\theta| = \frac{1}{p}\left(\frac{1}{q}\right)^{\log p/\log q} = \frac{1}{p^2} < 1.$$

So $F_\theta(x,y)$ is transcendental, and (3) makes A transcendental. The coefficient of A in (1) is $(q-p)/(q-1) \neq 0$ and the coefficient of A^2 in (2) is $-(p-1)/(q-1) \neq 0$, both rational. If either value were algebraic, its identity would exhibit A as a root of a nonzero polynomial over the algebraic numbers. □

The two identities are separately labelled because they carry different weight. A product of two transcendental numbers need not be transcendental, so the factorisation $\mathcal{R} = AC$ alone proves nothing; the quadratic identity (2) in the single value A is what settles the repeated sum. A third prime replaces the single Beatty boundary by a genuinely two-dimensional ordering problem, and the next section makes that failure exact.

4 The rank phase transition

One might hope to write the kernel as $f(i)g(j)h(k)$ and reduce the problem to one-dimensional criteria. At two generators that hope is exactly correct, and at three it fails at every finite order.

Proposition 4.1 (two generators separate)**.** *For $p,q > 1$ and all i,j, the two-prime kernel $\mathrm{K}_2(i,j) = 1/L_{p,q}(p^iq^j)$ is the outer product*

$$\mathrm{K}_2(i,j) = (p^iq^{\lfloor\log_q p^i\rfloor})^{-1}(p^{\lfloor\log_p q^j\rfloor}q^j)^{-1},$$

so every two-by-two minor of K_2 vanishes.

Proof. By the two-variable form of Theorem 2.1 the exponents of p and of q in $L_{p,q}(p^iq^j)$ are $i + \lfloor\log_p q^j\rfloor$ and $j + \lfloor\log_q p^i\rfloor$, each depending on one index only. A matrix whose entries are a product of a row function and a column function has vanishing two-by-two minors. □

Checked as the outer-product identity and the vanishing of every two-by-two minor. This is the linear-algebraic shadow of Section 3: the two-prime value factors because the kernel does.

At three generators the smallest rectangle already fails to factor. A factorisation $f(i)g(j)h(k)$ would force $\mathrm{K}(0,0,0)\,\mathrm{K}(1,1,0) = \mathrm{K}(1,0,0)\,\mathrm{K}(0,1,0)$.

Proposition 4.2 (non-separability at $\{2,3,5\}$)**.** *With* $(p,q,r) = (2,3,5)$*,*

$$\det\begin{pmatrix} \mathrm{K}(0,0,0) & \mathrm{K}(0,1,0) \\ \mathrm{K}(1,0,0) & \mathrm{K}(1,1,0)\end{pmatrix} = \det\begin{pmatrix} 1 & 1/6 \\ 1/2 & 1/60\end{pmatrix} = -\frac{1}{15} \neq 0.$$

Proof. The four values are computed from $\mathrm{H}(1) = 1$, $\mathrm{H}(2) = 2$, $\mathrm{H}(3) = 2\cdot 3 = 6$ and $\mathrm{H}(6) = 4\cdot 3\cdot 5 = 60$, so the determinant is $1/60 - 1/12 = -1/15$. □

Checked as the rank-two certificate and the non-separation witness, over the four exact values, the origin, value at two, value at three and value at six; the third-order witness $1/81000$ is checked beside the second-order one. The height at 6 is 60 because the maximal pure powers below 6 are 4, 3 and 5, so the running value at a smooth cutoff sees powers of the other primes that the cutoff itself omits.

Theorem 4.3 (arbitrary-order non-separability)**.** *Let p, q, r be primes with $p \neq q$, $p \neq r$ and $q \neq r$. For every $n \geq 1$ there are injective maps $I, J : \{0, \dots, n-1\} \to \mathbb{N}$ such that, for every $k \geq 0$,*

$$\det(\mathrm{K}(I(a), J(b), k))_{0\leq a,b<n} \neq 0.$$

Consequently, for no finite d do there exist rational-valued functions $f_\ell : \mathbb{N} \to \mathbb{Q}$ and $G_\ell : \mathbb{N}^2 \to \mathbb{Q}$, $0 \leq \ell < d$, satisfying $\mathrm{K}(i,j,k) = \sum_{\ell<d} f_\ell(i) G_\ell(j,k)$ for all i, j, k.

Proof. Put $\alpha = \log_r p$, $\beta = \log_r q$, $x_i = \{i\alpha\}$ and $y_j = \{j\beta\}$. The three height exponents are

$$\begin{aligned} v_p(\mathrm{H}(p^i q^j r^k)) &= i + \lfloor j\log_p q + k\log_p r\rfloor, \\ v_q(\mathrm{H}(p^i q^j r^k)) &= j + \lfloor i\log_q p + k\log_q r\rfloor, \\ v_r(\mathrm{H}(p^i q^j r^k)) &= k + \lfloor i\alpha\rfloor + \lfloor j\beta\rfloor + \lfloor x_i + y_j\rfloor. \end{aligned}$$

Every term except the last floor depends on i and k alone or on j and k alone, so with positive rational $R_i(k)$ and $C_j(k)$,

$$\mathrm{K}(i,j,k) = R_i(k)\, C_j(k)\, t^{\lfloor x_i + y_j\rfloor}, \qquad t = r^{-1}. \tag{4}$$

The remaining matrix is independent of k, and $x_i + y_j \in [0,2)$, so its entries are 1 and t.

Each of α and β is irrational, since a rational value would give an equality of positive powers of distinct primes, so each fractional-part orbit is dense in $(0,1)$ and each is injective. No simultaneous density of the pair of rotations is required. First choose indices with $0 < x_{I(0)} < \cdots < x_{I(n-1)} < 1$. For the columns write $s_b = 1 - y_{J(b)}$, and use density of the second orbit to choose

$$s_0 \in (0, x_{I(0)}), \qquad s_b \in (x_{I(b-1)}, x_{I(b)}) \quad (1 \leq b < n).$$

The intervals are disjoint, so the $J(b)$ are distinct. Their strict endpoints ensure that no selected entry lies on a threshold, and $x_{I(a)}+y_{J(b)} \geq 1$ holds exactly when $b \leq a$. Dividing row a by $R_{I(a)}(k)$ and column b by $C_{J(b)}(k)$ therefore leaves

$$C_n(t) = \begin{pmatrix} t & 1 & 1 & \cdots & 1 \\ t & t & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ t & t & \cdots & t & 1 \\ t & t & \cdots & t & t \end{pmatrix}, \qquad \det C_n(t) = t(t-1)^{n-1} \neq 0,$$

the determinant following by subtracting from each row its predecessor, working upwards from the last. Before the division, the determinant equals

$$\det C_n(t) \prod_{a<n} R_{I(a)}(k) \prod_{b<n} C_{J(b)}(k),$$

which is nonzero for every k; this also explains why the same indices work in every layer.

For the last assertion, suppose that a representation with d summands exists and fix k. On the selected rows $I(0), \ldots, I(d)$ and columns $J(0), \ldots, J(d)$ the resulting matrix factors as

$$(f_\ell(I(a)))_{0\leq a\leq d,\, \ell<d} (G_\ell(J(b),k))_{\ell<d,\, 0\leq b\leq d}.$$

It has rank at most d and hence zero determinant, contradicting the nonzero minor of order $d+1$. □

The factorisation (4) is the checked kernel factorisation, over the height factorisation. The uniform minor family is checked, the exclusion of every finite separation is checked, and the index selection is discharged by the problem-neutral staircase engine exists_-staircase_indices, which realises the full $n \times n$ pattern from two rotations without integer returns and depends on Mathlib only; the absence of integer returns is the checked irrationality of the two logarithm ratios. The Lean forms assume that p, q and r are prime with $p \neq r$ and $q \neq r$, and they do not use $p \neq q$. The proof uses the two one-dimensional density statements separately and needs no joint equidistribution hypothesis.

Index selection is essential. The leading minors of the $\{2,3,5\}$ kernel are not a witness: row three is $1/120$ times row zero for $j \leq 3$ (checked), and the proportionality fails at $j = 4$ (checked). A proof that read off leading minors alone would be false.

The same threshold description determines every finite sampled rank, rather than only producing one nonsingular minor.

Proposition 4.4 (finite sampled cut rank)**.** *Let $m \geq 1$ and let c lie in a field with $c \neq 0, 1$. For $0 \leq h \leq m$, let v_h be the length-m column whose first h entries are 1 and whose remaining entries are c. If the distinct columns of a matrix are the v_h with h in a nonempty set $E \subseteq \{0, \ldots, m\}$, then its rank is*

$$|E| - \mathbf{1}_{\{0,m\}\subseteq E}.$$

Proof. Write $E = \{h_1 < \cdots < h_s\}$. The $s-1$ consecutive differences are

$$v_{h_{a+1}} - v_{h_a} = (1-c)\mathbf{1}_{\{h_a,\ldots,h_{a+1}-1\}},$$

so they are linearly independent because their nonempty supports are disjoint. If $h_1 > 0$ or $h_s < m$, those supports miss a coordinate on which v_{h_1} is nonzero, and adjoining v_{h_1} gives rank s. If $h_1 = 0$ and $h_s = m$, the differences sum to $(1-c)\mathbf{1}$ while $v_0 = c\mathbf{1}$, so v_0 is already in their span and the rank is $s-1$. □

To apply the proposition, sort any chosen row phases x_i. Each normalised column is a threshold column v_h, where h is the number of sampled phases strictly below $1-y_j$. Repeated threshold positions give proportional columns before column normalisation, and the nonzero row and column factors in (4) do not change rank. Thus the displayed formula is the exact rank of every finite sample.

Attribution and reach. Fan recorded the qualitative obstruction for $|P| \geq 3$ in his comment of 26 June 2026 [12], observing that the two-prime argument does not seem to generalise immediately. The quantitative form above, its uniformity in the third coordinate, and its formalisation are proved here. Theorem 4.3 excludes exact separations of the displayed finite-sum form; it is not an independence, irrationality or transcendence statement, and it settles no instance of Erdős #269.

4.1 A sharp uniform obstruction for the normalised carry matrix

The determinant argument is exact and therefore fragile: an arbitrarily small perturbation of the kernel can destroy an exact vanishing. The next theorem replaces exactness by a uniform distance, and the constant it produces is sharp. It concerns the normalised carry matrix

$$C(i,j) = t^{\lfloor x_i + y_j \rfloor}, \qquad x_i = \{i \log_r p\}, \quad y_j = \{j \log_r q\}, \quad t = r^{-1}, \tag{5}$$

which is the factor left in (4) after the row and column factors are divided out. Say that a real matrix A on $\mathbb{N} \times \mathbb{N}$ has *finite separated rank* when all of its columns lie in one finite-dimensional space of real sequences, equivalently when $A(i,j) = \sum_{\ell<d} f_\ell(i) g_\ell(j)$ for some finite d and some sequences f_ℓ, g_ℓ, with no continuity or boundedness assumed.

Theorem 4.5 (sharp uniform separated approximation)**.** *Let p, q, r be pairwise distinct primes and let C be as in* (5)*. Then*

$$\inf_A \sup_{i,j \geq 0} |C(i,j) - A(i,j)| = \frac{1-t}{2} = \frac{r-1}{2r},$$

the infimum being over all matrices A of finite separated rank, and it is attained by the constant matrix of value $(1+t)/2$.

Proof. Every entry of C lies in $\{1, t\}$, and $C(i,j) = t$ exactly when $x_i + y_j \geq 1$. Fix $j \neq k$. The numbers y_j are pairwise distinct, since $\log_r q$ is irrational, so we may assume $y_j < y_k$, and then $0 \leq 1 - y_k < 1 - y_j \leq 1$. Density of the orbit (x_i) in $(0,1)$ supplies an index i with $1 - y_k < x_i < 1 - y_j$, and at that row the two columns carry the entries t and 1. Hence any two distinct columns of C are at sup-distance exactly $1-t$.

Let A have finite separated rank and put $\varepsilon = \sup_{i,j} |C(i,j) - A(i,j)|$. Suppose $\varepsilon < (1-t)/2$. Each column A_j then satisfies $\|A_j\|_\infty \leq 1 + \varepsilon$, so all columns of A lie in $W = V \cap \ell^\infty$, where V is the finite-dimensional space containing the columns of A; the space W is a subspace of V, hence finite-dimensional, and $\|\cdot\|_\infty$ is a genuine norm on it. By the triangle inequality, distinct columns satisfy

$$\|A_j - A_k\|_\infty \;\geq\; \|C_j - C_k\|_\infty - 2\varepsilon \;=\; 1 - t - 2\varepsilon \;>\; 0.$$

The compactness input is the standard fact that closed bounded subsets of a finite-dimensional real normed space are compact. The use of $W = V \cap \ell^\infty$ is essential: the individual separated row factors need not be bounded. The columns $A_j, j \geq 0$, are therefore infinitely many points of W, all of norm at most $1+\varepsilon$ and pairwise separated by a fixed positive distance. A bounded subset of a finite-dimensional normed space is totally bounded, so it contains no infinite uniformly separated family. This contradiction gives $\varepsilon \geq (1-t)/2$ for every A of finite separated rank.

For sharpness take $A(i,j) = (1+t)/2$, which has separated rank one; every entry of C is at distance exactly $(1-t)/2$ from it. □

The compactness step is standard, and the statement is recorded here for this kernel without a claim of technique. Three qualifications fix its reach. It is a statement about the normalised carry matrix (5): the original kernel carries positive row and column factors that decay, so the conclusion transfers to a weighted uniform norm relative to those factors and not to the unweighted sup norm of K. It bounds approximation of the whole infinite matrix, and on any finite range a separated approximant of small rank exists. And it says nothing about approximating the scalar sum, so it yields no irrationality conclusion. What it does add to Theorem 4.3 is robustness: no finite separated model of the carry, however chosen, gets uniformly closer than half a carry jump. The threshold is exact in both directions, since rank one attains it.

The original kernel behaves differently in the summation norm because its row and column factors decay. Let $K^{(N)}(i,j,k) = \mathrm{K}(i,j,k)$ for $i < N$ and $K^{(N)}(i,j,k) = 0$ otherwise. This is a sum of at most N terms separated between i and (j,k). Since $\mathrm{H}(x) > x^3/(pqr)$ for $x > 0$, geometric summation gives

$$\sum_{i,j,k\geq 0} |\,\mathrm{K}(i,j,k) - K^{(N)}(i,j,k)| \leq \frac{pqr\, p^{-3N}}{(1-p^{-3})(1-q^{-3})(1-r^{-3})}. \tag{6}$$

Hence the original summable kernel has finite separated-rank approximants in ℓ^1, even though its normalised carry matrix has the sharp uniform barrier of Theorem 4.5. This norm distinction is why neither statement decides the arithmetic nature of the scalar sum.

5 Dyadic blocks and the literal infinite tail

For the rest of the note set $P = \{2,3,5\}$ and $S = \mathcal{R}_P$, and write $\mathrm{H}_a = \mathrm{H}(2^a)$ and $h_a = \mathrm{H}_a/2$, so that $h_0 = 1/2$ while h_a is a positive integer for $a \geq 1$. Define the dyadic shell mass, its tail and its normalisation by

$$s_a = \sum_{\substack{i,j,k\geq 0\\ 2^a \leq 2^i 3^j 5^k < 2^{a+1}}} \frac{1}{\mathrm{H}(2^i 3^j 5^k)}, \qquad U_a = \sum_{j\geq a} s_j, \qquad \mathrm{X}_a = h_a U_a. \tag{7}$$

Compress the jump word of Section 2 into the blocks cut out by consecutive powers of two: block a starts just after 2^a, contains every pure 3- or 5-power strictly between 2^a and 2^{a+1}, and ends with the jump at 2^{a+1}. A channel cannot occur twice inside one block, because two powers of the same base $b \geq 2$ inside an interval of ratio $2 \leq b$ must coincide; this is the checked internal-power uniqueness lemma for the internal-power predicate. Let I_a list the internal jumps (p, e) with $p \in \{3, 5\}$ and $2^a < p^e < 2^{a+1}$, in increasing order.

Proposition 5.1 (the dyadic block alphabet)**.** *For every a,*

$$b_a = \frac{\mathrm{H}_{a+1}}{\mathrm{H}_a} = 2 \prod_{(p,e) \in I_a} p \in \{2, 6, 10, 30\}, \qquad \text{so} \qquad 2 \leq b_a \leq 30. \tag{8}$$

Proof. Internal-power uniqueness leaves two independent yes-or-no choices, one for the 3-channel and one for the 5-channel. Multiplying the terminal factor 2 by the selected channel factors gives exactly the four displayed cases. □

The definition is the dyadic block base, and Lean checks both the exact four-case alphabet and the bounded-radix consequence. All four letters occur early:

a	$(2^a, 2^{a+1})$	internal 3-power	internal 5-power	b_a
0	$(1, 2)$	none	none	2
1	$(2, 4)$	3	none	6
2	$(4, 8)$	none	5	10
3	$(8, 16)$	9	none	6
4	$(16, 32)$	27	25	30
5	$(32, 64)$	none	none	2

For $p \in P$ with q, r the other two primes, put

$$A_p(e) = \#\{(i, j) \in \mathbb{N}^2 : q^i r^j < p^e\}, \qquad C_p(e) = \sum_{u=1}^{e} A_p(u), \qquad C_p(0) = 0,$$

and for $(p, e) \in I_a$ let $\sigma_a(p, e)$ be the product of the channels of the later internal jumps of block a. The *ordered block digit* is

$$m_a = A_2(a+1) + \sum_{(p,e) \in I_a} (p-1)\, \sigma_a(p, e)\, (C_p(e) - C_2(a)) \quad (a \geq 1), \qquad m_0 = 1, \tag{9}$$

the integer implemented by the pinned checker and the checked ordered digit. The first four pairs (b_a, m_a) from $a = 1$ are $(6, 4)$, $(10, 7)$, $(6, 7)$, $(30, 65)$.

Theorem 5.2 (the literal shell recurrence)**.** *The shell masses are summable and $S = \sum_{a \geq 0} s_a$. For every $a \geq 0$,*

$$m_a = h_{a+1} s_a \in \mathbb{N}_{>0}, \qquad X_{a+1} = b_a X_a - m_a, \qquad X_a = \sum_{j \geq a} \frac{m_j}{b_a b_{a+1} \cdots b_j}. \tag{10}$$

Proof. Every exponent of 3 or 5 in the ath shell is at most a, and for each such pair Lemma 2.5 leaves at most one exponent of 2 placing the point in $[2^a, 2^{a+1})$. Each height is at least 2^a and the shell contains 2^a, so $0 < s_a \le (a+1)^2 2^{-a}$. The majorant is summable, which justifies every tail splitting below, and unique prime factorisation identifies $\sum_a s_a$ with the original repeated series.

Write the internal jumps as $t_\ell = p_\ell^{e_\ell}$, $1 \le \ell \le v$, and set $t_0 = 2^a$, $t_{v+1} = 2^{a+1}$. Let $N(t)$ count smooth positive integers strictly below t and put $n_\ell = N(t_\ell)$. On $[t_\ell, t_{\ell+1})$ the height is $\mathrm{H}_a \prod_{u \le \ell} p_u$, and the terminal jump has factor two, so

$$h_{a+1} s_a = \sum_{\ell=0}^{v} (n_{\ell+1} - n_\ell) \prod_{u>\ell} p_u = n_{v+1} - n_0 + \sum_{\ell=1}^{v} (p_\ell - 1)\Big(\prod_{u>\ell} p_u\Big)(n_\ell - n_0),$$

the second equality being finite summation by parts. Counting by the exponent of p gives $N(p^e) = C_p(e)$, so the right-hand side is exactly (9); at $a = 0$ the shell is the single point 1, giving $m_0 = 1$. Positivity follows from $s_a > 0$. Splitting $U_a = s_a + U_{a+1}$ and multiplying by h_a gives the recurrence, and $b_a \cdots b_j = \mathrm{H}_{j+1} / \mathrm{H}_a$ with $m_j = h_{j+1} s_j$ gives $m_j / (b_a \cdots b_j) = h_a s_j$, whose summation is the last identity. □

Summability is checked, and so is the recurrence for the genuine infinite tail, over the normalised state step.

Proposition 5.3 (integral state or cofinal separation)**.** *Either $X_a \in \mathbb{Z}$ for some $a \ge 0$, or for every a_0 there is $a \ge a_0$ with $|X_a - z| \ge 1/31$ for every $z \in \mathbb{Z}$.*

Proof. Suppose cofinal separation fails, so for some A and every $a \ge A$ there is an integer z_a with $e_a = X_a - z_a$ and $|e_a| < 1/31$. The recurrence gives $z_{a+1} - b_a z_a + m_a = b_a e_a - e_{a+1}$. The left side is an integer and the right side has absolute value below $(30+1)/31 = 1$, so both vanish and $e_{a+1} = b_a e_a$. Hence $|e_{A+k}| \ge 2^k |e_A|$ for every k while $|e_{A+k}| < 1/31$, which forces $e_A = 0$. □

This is checked for the genuine infinite tail, over the abstract integral-state or cofinal-distance alternative. It excludes neither branch, and it says nothing about a rational value whose reduced denominator has a factor coprime to 30: that case is the subject of Section 6.

6 A quadratic tail bound and denominator cancellation

Put

$$n_a = a + \lfloor \log_3(2^a) \rfloor + \lfloor \log_5(2^a) \rfloor, \qquad Q(n) = \frac{n^2 + 8n + 18}{9},$$

so n_a is the sum of the three height exponents at 2^a.

Theorem 6.1 (quadratic bound for the actual tail)**.** *For every $a \ge 0$, $0 < X_a \le Q(n_a)$.*

Proof. Partition the smooth integers $x \ge 2^a$ by their height vector $(A, B, C) = (\lfloor \log_2 x \rfloor, \lfloor \log_3 x \rfloor, \lfloor \log_5 x \rfloor)$. The cell of a vector is the interval $[\lambda, \eta)$ with $\lambda = \max(2^A, 3^B, 5^C)$ and $\eta = \min(2^{A+1}, 3^{B+1}, 5^{C+1})$, so $\eta \le 2^{A+1} \le 2\lambda$ and, by Lemma 2.5, fixing the exponents of 3 and 5 leaves at most one exponent of 2. Since $A \geq B \geq C$, the cell contains at most

$$(B+1)(C+1) \leq \frac{(A+B+C+3)^2}{9}$$

smooth points: writing $u = B+1 \geq v = C+1$ and using $A+1 \geq u$, the difference $(2u+v)^2 - 9uv = (u-v)(4u-v)$ is nonnegative. Unique factorisation identifies these exponent triples with distinct smooth integers.

As the cutoff increases all three height exponents are nondecreasing, so two nonempty cells with the same exponent sum are the same cell, and there is at most one nonempty cell of each rank. Every cell above 2^a has rank at least n_a, and a cell of rank $n_a + k$ has height at least $\mathrm{H}_a\, 2^k$, since each of the k extra prime factors is at least 2. Nonnegative summation over ranks, allowing empty ones, gives

$$X_a \leq \frac{1}{18} \sum_{k \geq 0} \frac{(n_a + k + 3)^2}{2^k} = \frac{n_a^2 + 8n_a + 18}{9},$$

the evaluation using the geometric moments $\sum 2^{-k} = 2$, $\sum k 2^{-k} = 2$ and $\sum k^2 2^{-k} = 6$. Positivity follows from the shell at 2^a. ☐

The rank here is the sum of the three fixed height exponents, not a varying-smoothness density parameter. The estimate uses only unique factorisation, the short-cell counting bound and geometric moments; it does not invoke a Dickman–Hildebrand asymptotic or a Hecke–Mahler theorem.

This is the analytic content of the section: each additional height rank costs a geometric factor while its multiplicity grows only quadratically. The finite projection and the sorted quadratic inequality are the checked ingredients of Proposition 2.6; grouping the actual infinite tail by height cells and summing the majorant is the argument above.

The bound is used at the endpoint of a window, where the natural index is the positive prime-power jump count strictly below the cutoff. Put

$$j_a = \#\{p^e < 2^a : p \in \{2,3,5\},\ e \geq 1\} = n_a - 1, \tag{11}$$

the equality holding because the powers of 2 below 2^a number $a-1$ while the powers of 3 and of 5 below 2^a number $\lfloor \log_3 2^a \rfloor$ and $\lfloor \log_5 2^a \rfloor$. Substituting $n_a = j_a + 1$ into Q gives the integer bound used throughout the rest of the note:

$$K(B,a) = \lfloor B\, Q(n_a) \rfloor = \left\lfloor \frac{B(j_a^2 + 10j_a + 27)}{9} \right\rfloor. \tag{12}$$

The cutoff in (11) is 2^a and it is strict.

Lemma 6.2 (finite denominator clearing). *For all integers $0 \leq u \leq b$ the window mass $h_b \sum_{a=u}^{b-1} s_a$ is a natural number. If $S = N/D$ with $N \in \mathbb{Z}$ and $D \in \mathbb{N}_{>0}$, then $DX_a \in \mathbb{Z}$ for every $a \geq 1$, and there are indices $1 \leq i < j \leq D+1$ for which $X_i - X_j \in \mathbb{Z}$.*

Proof. An empty window has mass zero. Otherwise $b \geq 1$, and every integer $x < 2^b$ has 2-height exponent at most $b-1$ while its other height exponents are at most those at 2^b, so $\mathrm{H}(x) \mid \mathrm{H}_b/2 = h_b$ and every term of the finite window clears at h_b. Writing $v_a = h_a \sum_{u<a} s_u \in \mathbb{N}$ gives $DX_a = h_a N - Dv_a \in \mathbb{Z}$. Among $D+1$ of these integers two share a residue modulo D, and the corresponding states differ by an integer. □

The strict upper endpoint is what permits the division by two. Checked as the window clearing identity, the all-scale lattice and the collision, over the boundary divisibility and the half-height identity.

Theorem 6.3 (positive reduced carries from rationality). *Suppose $S = N/D$ with $N \in \mathbb{Z}$, $D \in \mathbb{N}_{>0}$, and write*

$$D = 2^u 3^v 5^w B, \qquad u, v, w \in \mathbb{N}, \quad B \in \mathbb{N}_{>0}, \quad \gcd(B, 30) = 1, \qquad a_D = u + 1 + 2v + 3w.$$

Then for every $a \geq a_D$ the number $z_a = BX_a$ is a positive integer and

$$z_{a+1} = b_a z_a - Bm_a, \qquad 1 \leq z_a \leq K(B, a) \leq 90B(a+1)^2.$$

Proof. Let $M = 2^u 3^v 5^w$. For $a \geq a_D$ we have $2^a \geq 2^{u+1}$, $2^a \geq 3^v$ and $2^a \geq 5^w$, using $3 < 2^2$ and $5 < 2^3$; hence $M \mid h_a$. In the clearing identity $DX_a = h_a N - Dv_a$ of Lemma 6.2 both terms on the right are divisible by M, so dividing by M shows that BX_a is an integer. Positivity and the recurrence come from (10), and the upper bound is Theorem 6.1 with the floor taken, since z_a is an integer below $BQ(n_a)$. Finally $n_a \leq 3a$, so $Q(n_a) \leq a^2 + \frac{8}{3}a + 2 \leq 90(a+1)^2$. □

The smooth part M affects only the onset a_D; the surviving carry bound depends on the coprime factor B. The bridge is checked against the cruder width $90B(a+1)^2$, which is the form the formal consumer uses; the particular onset $a_D = u + 1 + 2v + 3w$ and the sharper bound $K(B, a)$ are the paper statement above. The abstract cancellation is checked separately: every fixed smooth factor divides the running height once the cut-off reaches it (absorption), a shared factor cancels from the recurrence (cancellation), and positivity, the bound and the window identity descend through it (bound transfer, window transfer).

The integral branch has one further exact property, recorded because it constrains any attempt to build a surviving seed by hand.

Proposition 6.4 (upward closure and rigidity). *For every a, $X_a = (m_a + X_{a+1})/b_a > 0$, and if $X_a \in \mathbb{Z}$ then $X_n \in \mathbb{Z}$ for every $n \geq a$. Moreover, fix A, a positive width function w with $w(A+k)/2^k \to 0$, and a real orbit $(y_n)_{n \geq A}$ satisfying $y_{n+1} = b_n y_n - m_n$. If y_n and X_n both lie in $(m_n/b_n,\ m_n/b_n + w(n)]$ for every $n \geq A$, then $y_A = X_A$.*

Proof. The identity is the recurrence solved for X_a, and positivity holds because every shell contains its dyadic left endpoint. Integer coefficients give upward closure. For the last assertion, the difference of the two orbits is multiplied by $b_n \geq 2$ at each step while the common window bounds its absolute value at time $A + k$ by $w(A+k)$, so $2^k |y_A - X_A| \leq w(A+k)$ and the stated decay forces equality. □

7 Residue windows and the equivalence band

The statements of this section are about integer sequences. For a start $\ell \geq 1$ and a length $h \geq 1$ define

$$W_{\ell,h} = \prod_{j=0}^{h-1} b_{\ell+j}, \qquad F_{\ell,0} = 0, \qquad F_{\ell,h+1} = b_{\ell+h}F_{\ell,h} + m_{\ell+h},$$

the window base and the window forcing. Iterating the recurrence gives the division-free identity

$$X_{\ell+h} = W_{\ell,h}X_\ell - F_{\ell,h}, \qquad \text{and} \qquad z_{\ell+h} = W_{\ell,h}z_\ell - BF_{\ell,h} \tag{13}$$

for any integral carry with $z_{n+1} = b_n z_n - Bm_n$; the integral form is the checked window identity, over the affine window identity and the scaled forcing identity, and the real form is checked for the genuine tail. For $C > 0$ let $\mathrm{lpr}_C(N)$ be the unique integer in $\{1, \ldots, C\}$ congruent to N modulo C, so $\mathrm{lpr}_C(0) = C$; this is the canonical representative, with its range and its congruence checked. The vanishing residue is represented by C, and that choice is what makes the next statement a modular one.

Proposition 7.1 (the finite residue contradiction). *Let $C > 0$ and let c be an integer with $0 < c$ and $|c| \leq K$. If $c \equiv N \pmod{C}$ and $K < \mathrm{lpr}_C(N)$, then the hypotheses are contradictory.*

Proof. The canonical representative lies in $\{1, \ldots, C\}$, and the inequalities put c strictly between 0 and C. If $N \equiv 0 \pmod{C}$ its representative is C while $c \bmod C = c \neq 0$. Otherwise c and $\mathrm{lpr}_C(N)$ are each their own residue and congruence makes them equal, contradicting $|c| \leq K < \mathrm{lpr}_C(N)$. □

The abstract one-window predicate is the residue-escape condition, with the contrapositive bounding the residue from a bounded positive state. The natural-state version is checked, the integer carry version is checked, and the exact classifier $\mathrm{lpr}_C(N) = |c|$ under $|c| \leq C$ is checked as well. Applying it inside a window uses the forcing of the endpoint carry to the canonical residue, checked.

For a bound $G : \mathbb{N}_{>0} \times \mathbb{N} \to \mathbb{N}$ let $\mathsf{E}(G)$ be the statement

$$\begin{gathered}\text{for every } B \geq 1 \text{ with } \gcd(B, 30) = 1 \text{ and every } a_0 \geq 1,\\ \text{there are } \ell \geq a_0 \text{ and } h \geq 1 \text{ with } \mathrm{lpr}_{W_{\ell,h}}(-BF_{\ell,h}) > G(B, \ell + h).\end{gathered} \tag{14}$$

The window may depend on both B and a_0. The general predicate is the cofinal local-window escape condition, and the instance at the fixed bound $K_0(B, a) = 90B(a+1)^2$ is the actual producer. Every b_a is positive, so $W_{\ell,h} > 0$ is automatic.

Theorem 7.2 (the equivalence band). *Let $G : \mathbb{N}_{>0} \times \mathbb{N} \to \mathbb{N}$ satisfy $K(B, a) \leq G(B, a)$ for all B and a, and $G(B, a)/2^a \to 0$ as $a \to \infty$ for each fixed B. Then*

$$\mathsf{E}(G) \iff S \notin \mathbb{Q}.$$

Both K of (12) and $K_0(B, a) = 90B(a+1)^2$ lie in this band, so $\mathsf{E}(K)$, $\mathsf{E}(K_0)$ and irrationality of S are mutually equivalent. The lower half of the band cannot be dropped: $\mathsf{E}(0)$ holds vacuously, since every least positive residue is at least 1.

Proof. Suppose $\mathsf{E}(G)$ and suppose $S = N/D$ were rational. Theorem 6.3 supplies $B \geq 1$ coprime to 30, an onset a_D, and positive integers $z_a = BX_a \leq K(B,a) \leq G(B,a)$ for $a \geq a_D$ satisfying the cleared recurrence. Apply (14) with $a_0 = a_D$ to obtain a window (ℓ, h) with $\ell \geq a_D$. By (13), $z_{\ell+h} \equiv -BF_{\ell,h}$ modulo $W_{\ell,h}$, while $0 < z_{\ell+h} \leq G(B, \ell+h) < \mathrm{lpr}_{W_{\ell,h}}(-BF_{\ell,h})$. Proposition 7.1 is the contradiction, so S is irrational.

Conversely suppose $S \notin \mathbb{Q}$, and fix $B \geq 1$ coprime to 30 and $a_0 \geq 1$. Set $\ell = a_0$. In $X_\ell = h_\ell(S - \sum_{a<\ell} s_a)$ the finite prefix is rational and h_ℓ is a positive rational, so BX_ℓ is irrational and its distance δ from $\mathbb{Z}$ is positive. Suppose no length h escaped, so that $r_h = \mathrm{lpr}_{W_{\ell,h}}(-BF_{\ell,h}) \leq G(B, \ell+h)$ for every $h \geq 1$. Then $k_h = (BF_{\ell,h} + r_h)/W_{\ell,h}$ is an integer, and multiplying (13) by B gives

$$BX_\ell - k_h = \frac{BX_{\ell+h} - r_h}{W_{\ell,h}}.$$

Both $BX_{\ell+h}$ and r_h are positive, the first at most $BQ(n_{\ell+h})$ by Theorem 6.1 and the second at most $G(B, \ell+h)$, so the numerator has absolute value at most $\max(BQ(n_{\ell+h}), G(B, \ell+h))$. Each $b_a \geq 2$ gives $W_{\ell,h} \geq 2^h$, so

$$\begin{aligned} 0 < \delta \leq |BX_\ell - k_h| &\leq \frac{\max(BQ(n_{\ell+h}),\, G(B,\ell+h))}{2^h} \\ &= 2^\ell \cdot \frac{\max(BQ(n_{\ell+h}),\, G(B,\ell+h))}{2^{\ell+h}} \longrightarrow 0. \end{aligned}$$

as $h \to \infty$, because $Q(n_{\ell+h})$ is quadratic in $\ell + h$ and $G(B,a) = o(2^a)$. This contradiction proves $\mathsf{E}(G)$.

For the two named bounds, $K \leq K_0$ by Theorem 6.3, both are quadratic in a for fixed B, and $K \leq K$ trivially. Finally $\mathrm{lpr}_C(N) \geq 1 > 0$ always, so $\mathsf{E}(0)$ holds whatever S is, while $0 \leq K$ fails. □

The converse direction is checked in Lean at a strictly greater generality than the fixed bound: irrationality implies escape against any short bound that the window growth eventually beats, checked, and in particular against every bound dominated by $c(B)(n+1)^2$, checked. The enlarged little-$o(8^a)$ band, with the cap-domination hypothesis retained, is checked by the octic window-band equivalence. The forward direction at the fixed bound K_0 is checked, over the abstract consumer checked and its packaged absorbed form checked. The two directions combine into the equivalence checked for the shifted tail and checked for the series value itself. The Lean equivalence is stated at K_0; the band of Theorem 7.2 is the paper statement.

Theorem 7.2 settles a question that a reader is entitled to ask about the criterion. Sharpening the bound below K does not weaken the producer, since escape against a bound that fails to dominate the actual carries proves nothing; the zero bound is the extreme case. Enlarging the bound up to any subexponential function does not weaken it either. The producer is therefore a restatement of Erdős #269 for $P = \{2, 3, 5\}$ throughout the band, and work on it is work on the target. Equivalence preserves truth and settles nothing about difficulty, so the reformulation may still be the easier representation to attack; what it does not admit is a cheaper bound.

Two exact countermodels rule out further shortcuts. For $(W, F, B) = (6, 4, 1)$ we have $\mathrm{lpr}_6(-4) = 2$, so the canonical residue need not be coprime to the accumulated base. For $(W, F, B) = (60, 47, 37)$ we have $\mathrm{lpr}_{60}(-37 \cdot 47) = 1$, so a fixed window has no denominator-independent lower bound on that residue. The window may therefore depend genuinely on B.

7.1 How fast the window base grows

The crude estimate $W_{\ell,h} \geq 2^h$ is all the equivalence needs. The actual height formula gives more, and the extra information sets the scale of any finite search.

Proposition 7.3 (window-growth law)**.** *Put $\theta_3 = \log_3 2$ and $\theta_5 = \log_5 2$. For all $\ell \geq 0$ and $h \geq 1$,*

$$W_{\ell,h} = 2^h\, 3^{\lfloor(\ell+h)\theta_3\rfloor - \lfloor \ell\theta_3\rfloor}\, 5^{\lfloor(\ell+h)\theta_5\rfloor - \lfloor \ell\theta_5\rfloor}, \qquad \frac{8^h}{15} < W_{\ell,h} < 15 \cdot 8^h.$$

Proof. Telescoping (8) gives $W_{\ell,h} = \mathrm{H}_{\ell+h} \,/\, \mathrm{H}_\ell$, and the displayed formula is that quotient written out. Each floor difference differs from $h\theta_p$ by less than one, and $3^{\theta_3} = 5^{\theta_5} = 2$, so the 3-factor lies strictly between $2^h/3$ and $3 \cdot 2^h$ and the 5-factor strictly between $2^h/5$ and $5 \cdot 2^h$. Multiplying the three ranges gives the bounds. □

Corollary 7.4 (no bounded-length escape at all starts)**.** *Fix $B \geq 1$ coprime to 30 and $H \geq 1$. Only finitely many starts ℓ admit an escaping window of length at most H against the bound K.*

Proof. A least positive residue never exceeds its modulus, so escape at (ℓ, h) requires $K(B, \ell + h) < W_{\ell,h} < 15 \cdot 8^h \leq 15 \cdot 8^H$. On the other hand $j_a \geq a - 1$, since the powers $2, \ldots, 2^{a-1}$ already lie below 2^a, so $K(B, \ell + h) \geq \lfloor B((\ell - 1)^2 + 10(\ell - 1) + 27)/9 \rfloor$, which tends to infinity with ℓ. □

Corollary 7.4 is the exact reason a finite scan cannot approach the cofinal quantifier by widening its denominator range alone. Rearranging its inequality, an escaping window at start ℓ and endpoint $a = \ell + h$ must satisfy

$$3h > \log_2 K(B, a) - \log_2 15,$$

so the necessary search depth grows like $(\log_2 B + 2\log_2 \ell)/3$. This is a lower bound on the length that can possibly work, and it is not an upper bound on the first length that does. Producing the latter is exactly the open problem, and by Theorem 7.2 it needs effective control of $\mathrm{dist}(BX_\ell, \mathbb{Z})$: counting window growth is easy, and keeping a scaled tail away from the integers is the arithmetic content.

Status. The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no `sorry`, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

Companion system context. The claim and trust boundary, cold-clone route to proof authority, and public contribution protocol are described in sibling papers. Those descriptions do not change the mathematical status of this note.

8 The finite evidence

Three finite computations bear on the problem. Each is exact integer or certified-enclosure arithmetic, none is a Lean theorem, and none settles an instance.

8.1 The dyadic-window scan

A *certificate* is a finite tuple of integers recording one instance of the inequality in (14), so that a reader can recheck it in a line. The integer-only dyadic-window checker constructs the ordered pure-power jumps, the block bases and the block digits from exact multiplicity counts, and reproduces the following, with columns the denominator B, the start ℓ, the length h, the endpoint jump index $j_{\ell+h}$, the window base, the forcing, the residue $R = \mathrm{lpr}_W(-BF)$ and the bound K of (12).

B	ℓ	h	$j_{\ell+h}$	W	F	R	K
1	1	2	4	60	47	13	9
7	1	3	5	360	289	137	95
16	1	4	7	10800	8735	640	352

The first row reads as follows. The window starts at $\ell = 1$ and has length 2, so $W = b_1 b_2 = 6 \cdot 10 = 60$; the accumulated forcing is $F = 47$; and $\mathrm{lpr}_{60}(-47) = 13$, since $-47 + 60 = 13$, which exceeds $K(1,3) = \lfloor (16 + 40 + 27)/9 \rfloor = 9$. The third row lies outside the domain of (14), since $\gcd(16, 30) = 2$, and is displayed to illustrate the window arithmetic at greater depth.

A fresh scan over every $B \le 5000$ coprime to 30 and every $100 \le \ell \le 3000$ tested 3,869,934 pairs and found an escaping window in every case, of length at most 18 with search depth permitted to 24. The distribution of first successful lengths concentrates at 10 to 12, which is where Corollary 7.4 predicts the shortest possible window to lie over this range. The computation uses integers only and is reproducible from the pinned checker with `--max-denominator 5000 --start-min 100 --start-max 3000 --max-length 24 --assert-packet`. Neither the scan nor the three displayed certificates proves escape for unbounded B or for cofinally many starts, and by Corollary 7.4 no scan at bounded length can.

8.2 Two finite denominator exclusions

Theorem 8.1 (window-128 block exclusion, computational certificate)**.** *At window length $L = 128$, launch $a_1 = 10005$ and 64 starts, no rational value of the $\{2, 3, 5\}$ running-LCM series has reduced denominator MB with M a 30-smooth divisor of $2^{10005}3^{6312}5^{4308}$, $\gcd(B, 30) = 1$ and $1 < B \le B_{\max}$, where $B_{\max}$ is the 106-digit integer*

$$B_{\max} = 1134599670999687767349520845707093359257353022286558739363600235\\ 0161032075640633732703053241721452819717 29,$$

so that $\log_2 B_{\max} = 348.9846\ldots$

The exponent triple $(10005, 6312, 4308)$ is the height exponent triple of $\mathrm{H}(2^{10005})$, so the certificate normalises by the full height at its launch. Lemma 6.2 normalises by the half height h_{10005}; the certificate's smooth family is accordingly the larger one. The recorded quantities are the window product with $\log_2 P = 386.40993\ldots$, the exclusion index $J = 1$ certified from a reduced basis lying inside the per-start budget, an enclosure of width 9.674×10^{-227}, and a maximum ratio $\max X/W = 0.185997$. This is a computational certificate and nothing else: no Lean declaration carries any part of it, so it rests on the correctness of the exact integer engine that produced it, and the certificate record is held in the author's formal-mathematics archive, outside the public source of this release.

Theorem 8.2 (continued-fraction exclusion, computational certificate)**.** *The normalised tail X_1 has* 13,109 *certified partial quotients, so it is not rational with denominator at most* 2^{22482}*, about* 10^{6768}*.*

The certification is by common prefix of the continued fractions of the two endpoints of an interval provably containing X_1, so no approximation heuristic enters, and the truncation was independently checked to agree with the direct smooth-number sum as an exact rational. The machine-readable witness and replay receipt for this large computation are not included in the public release, so the statement remains an archived ordinary computational certificate rather than a publicly replayed or Lean-checked result. The recorded statistics are a largest denominator of 22,483 bits, a largest partial quotient of 129,114, a mean partial quotient of 23.4133, observed Gauss–Kuzmin frequencies 0.4208, 0.1665, 0.0917, 0.0575, 0.0391 against the predicted 0.4150, 0.1699, 0.0931, 0.0589, 0.0406, and a Lévy constant of 1.18869 against $\pi^2/(12\log 2) = 1.18657$. No Liouville behaviour and no algebraic or self-similar continued-fraction structure appears below that height.

Each exclusion is finite and neither contains the other: one is indexed by a lattice at a fixed launch and fixed smooth part, the other by continued-fraction depth at $a = 1$. Denominators beyond the recorded finite ranges are not excluded by either certificate, so neither settles an instance of the problem.

9 The remaining arithmetic questions

Problem 9.1 (actual cofinal residue escape)**.** Prove $\mathsf{E}(K)$ for the actual digits (9) and the bound (12).

Every term in this question is a finite integer quantity, and a proof must produce a later window for every reduced denominator and every prescribed onset. The tail estimate and the smooth-factor cancellation are proved above and are not additional hypotheses. By Theorem 7.2 the problem is a restatement of Erdős #269 for $P = \{2, 3, 5\}$.

Problem 9.2 (nonintegrality of every reduced tail)**.** For every $B \geq 1$ with $\gcd(B, 30) = 1$ and every $a \geq 1$, prove

$$BX_a \notin \mathbb{Z}. \tag{15}$$

Proposition 9.3 (the reduced-tail question is also the target)**.** *Statement* (15)*, quantified over every $B \geq 1$ coprime to* 30 *and every $a \geq 1$, is equivalent to irrationality of S.*

Proof. If $BX_a \in \mathbb{Z}$ for some such B and a, then $X_a \in \mathbb{Q}$, and since $S = \sum_{j<a} s_j + X_a/h_a$ with h_a a positive rational and the prefix a finite sum of rationals, $S \in \mathbb{Q}$. Conversely if $S \in \mathbb{Q}$, then Theorem 6.3 produces B coprime to 30 with $BX_a \in \mathbb{Z}$ for every $a \geq a_D$. □

The abstract form of the first implication is checked. Problem 9.2 is therefore a second restatement of the target, and its pointwise shape is the more convenient one: if BX_a is integral at one index, the integer-coefficient recurrence makes it integral at every later index, so a single index decides it. Proposition 5.3 does not exclude that branch, since it constrains X_a alone and a rational value may keep a surviving denominator coprime to 30.

9.1 The analytic route and what it would need

For the de-duplicated series put

$$\mathcal{D}_{2,3,5} = 1 + \sum_{t \in \{2^n, 3^n, 5^n : n \geq 1\}} \frac{1}{2^{\lfloor \log_2 t \rfloor} 3^{\lfloor \log_3 t \rfloor} 5^{\lfloor \log_5 t \rfloor}},$$

whose dyadic coding is the joint rotation word $\delta_{3,a} = \lfloor (a+1)\theta_3 \rfloor - \lfloor a\theta_3 \rfloor$ and $\delta_{5,a} = \lfloor (a+1)\theta_5 \rfloor - \lfloor a\theta_5 \rfloor$, with $b_a = 2 \cdot 3^{\delta_{3,a}} 5^{\delta_{5,a}}$.

Problem 9.4 (function-faithful two-dimensional representation)**.** Express $\mathcal{D}_{2,3,5}$ as a nonconstant algebraic combination of values of a specified two-dimensional Hecke–Mahler, cone-generating or multivariate Mahler function and verify every hypothesis of a published value theorem; or give a conditional theorem under an explicit logarithmic nondegeneracy hypothesis; or prove that the literal series has no representation in the specified finite-dimensional class.

Individual irrationality of θ_3 and θ_5 is what Theorem 4.3 uses, and it is weaker than the joint equidistribution or the rational independence of $1, \theta_3, \theta_5$ that a two-dimensional value theorem may need; the stronger hypothesis is not assumed anywhere above. The finite-observer formalisation isolates the precise faithfulness requirement: equality in a finite observer must imply equality after symbolic realisation, and a genuine finite-dimensional factorisation forces the realised symbolic span to be finite-dimensional (residue-coboundary form, symbolic realisation, checked), with the carry residue and residue digit confined to their declared intervals (carry interval, digit interval). No theorem here proves that the literal realised span is infinite.

9.2 Structural constraints already in place

The radix alphabet extends without difficulty. For ordered primes $p_1 < \cdots < p_s$ an interval (p_1^a, p_1^{a+1}) contains at most one power from each other channel, because consecutive p_i-powers have ratio $p_i > p_1$, so its block radix belongs to the 2^{s-1}-letter alphabet $\{p_1 \prod_{i=2}^{s} p_i^{\varepsilon_i} : \varepsilon_i \in \{0,1\}\}$. The quantitative questions are the interesting ones: effective recurrence or discrepancy for the actual four-letter $\{2, 6, 10, 30\}$ word, an asymptotic with an error term for the restricted two-dimensional shell counts that generate m_a, or the exact separated rank of the literal kernel under a specified family of shifts.

Three-channel rigidity and carry-lift extinction already exclude one proposed argument in four exact steps. Under channel surjectivity, ordinary block nullity is equivalent to the perturbation being a coboundary of a channel potential (potential classifier). Zero perturbations on genuine $2 \to 3$ and $2 \to 5$ transitions then identify all three potential values and force the perturbation to vanish at every index. For an integral lift, that vanishing makes a nonzero initial error grow by the exact product of the successive bases, and bases at least two make its absolute value at least 2^N, contradicting even a single index-N bound strictly below 2^N (one-index extinction); consequently no uniform bound on the lift error can hold either (uniform extinction). A separate four-state calculation reaches the obstruction earlier: four real states in $(0, 1)$ with unit-accuracy integral lifts and the two anchor equalities force the first complete 2-block sum to be 1, so that block cannot be null (first-block sum, four-state obstruction).

These lift conclusions are conditional. The paper constructs the actual orbit and its bounded integral carry under rationality, and it does not construct a faithful lift with the two anchors or the block nullity those auxiliary statements require. The actual carry supplies a weighted block defect instead, so any successful argument along that line must use the weighted identity or construct a different faithful lift.

A single conditional single-channel criterion is also on record. For the pure 2-channel sum the Cantor states lie in an explicit open interval (confinement), consecutive states are separated (gap bound), a small nonzero linear form forces irrationality (criterion), and clearing together with small gaps assembles to irrationality (assembly). The clearing and small-gap hypotheses are not discharged for any concrete series, so no irrationality statement follows from them here.

9.3 Where the problem stands

Erdős #269 is settled for $|P| = 2$, at the level of transcendence, and open for every $|P| \geq 3$. For $P = \{2, 3, 5\}$ the reduction is complete on the arithmetic side: the literal digit, the tail, the integral recurrence, the smooth-factor cancellation, the explicit onset and the sharp carry bound all refer to the same series, and Theorem 7.2 shows that the one remaining condition is the problem itself throughout an entire band of bounds. Three obstructions mark the limits. Arbitrary-order minors exclude every finite exact separation of the kernel, and Theorem 4.5 excludes even approximate separation of its normalised carry below half a jump. The bounded-radix alternative leaves integral tails untouched. The residue countermodels show that coprimality alone gives no denominator-independent residue bound, and Corollary 7.4 shows that no bounded-length search can reach the cofinal quantifier. What remains is to force the literal word's residue outside the finite carry interval, with the window allowed to depend on the denominator and to begin beyond the onset.

Statements and declarations

Evidence. Steve Fan's two-prime argument is an ordinary proof over the cited Bugeaud–Laurent theorem and carries no Lean declaration. The arbitrary-order kernel minors and the exclusion of finite separation are Lean theorems in `Kernel-CarryRank`. The named source statements for the literal recurrence and fixed-split rationality bridge are Lean theorems as well. The source-current validation receipt byte-matches these modules and records their compilation inside the public recursive import closure; its named declaration-by-declaration axiom audit is for the R12 Wave A results, not for these rank declarations. Comparator names the finite $(2, 3, 5)$ minor $-1/15$, the running-LCM height identity, and a conditional carry-escape consumer; it is not the evidence for the arbitrary-order theorem. The actual-series reduction is given; the source-specific cofinal escape remains unproved. Named Lean sources below are locators, not a blanket end-to-end coverage claim. The quadratic tail bound, the explicit onset a_D, the sharp bound K, and Theorem 4.5 are ordinary proofs in this paper; the three finite computations of Section 8 are computational certificates.

Artefact and data availability. The pinned formal-source revision contains the Lean sources, the fixed toolchain, the library manifest, and the exact dyadic-window checker used in the finite scan. Proof authority rests in those pinned sources. The present text is exposition and navigation.

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Acknowledgements. The two-prime factorisation, its Hecke–Mahler reduction and the running-LCM identity for every $|P|$ are credited to Steve Fan. The transcendence input is credited to Yann Bugeaud and Michel Laurent and to the earlier work of Loxton and van der Poorten cited by them. The problem numbering and status follow the Erdős Problems catalogue maintained by Thomas Bloom [4].

10 Extended record: migrated arguments and finite evidence

This part of the record holds material that the short note compressed or moved. Nothing here is an additional result; every statement is either an auxiliary form of a theorem of the note, a historical or bibliographic detail, or the exact source inventory. The canonical references are the note's own labels.

10.1 The historical and catalogue record

In the primary 1988 source Erdős states the infinite-P assertion as a simple exercise and presents persistence for a finite number of primes greater than one as a probable extension, and not as a theorem [2, p. 106]. The current open status is supplied by the catalogue [4], and is not inferred from that conjectural wording. The 1974 letter writes "given primes $p_1, \ldots, p_r$" without explicitly restricting r, so the singleton case, whose value is $p/(p-1)$, must be excluded by hand; the modern restriction $|P| \geq 2$ does that.

The letter's de-duplicated assertion is made for a general finite list of primes and supplies no proof, so it is an asserted historical result and not an open problem. The note therefore restricts its open statement to the repeated series $\mathcal{R}_P$. A recovered proof of the letter's assertion, or a transcendence statement for $\mathcal{D}_P$ with $|P| \geq 3$, would be a different question and needs its own formulation.

The two-prime argument of the note is independent of the unprinted argument in the letter, and it is not the first public proof. Steve Fan posted the same factorisation, the same Hecke–Mahler reduction and the same conclusion in the discussion thread of the problem's page on 26 June 2026 [12], with follow-up remarks extending the argument to arbitrary coprime pairs; a further comment there observes that the argument does not seem to generalise immediately to $|P| \geq 3$. The manuscript of the note was first released publicly on 22 July 2026, at commit `a9d3ab8`.

The statement of the problem has been formalised as a conjecture with an unfilled proof in the *Formal Conjectures* collection [5]. Its Nat-indexed series includes the empty-prefix least-common-multiple term, so its value differs from the conventional one by a rational constant; transporting a theorem across that boundary needs an explicit series-identification lemma, which is not supplied here.

10.2 The abstract carry lemmas in their general form

The note applies the carry machinery to the literal $\{2, 3, 5\}$ orbit. The underlying statements are about integer sequences alone, they are reusable, and they are recorded here in the generality in which they are checked.

Let $D = D_{\mathrm{sm}} B$ with $D_{\mathrm{sm}} = 2^u 3^v 5^w$ and $\gcd(B, 30) = 1$, and let (c_n) be an integer sequence satisfying $c_{n+1} = b_n c_n - D m_n$ for an integer radix word (b_n) and forcing word (m_n).

Proposition 10.1 (conditional denominator reduction)**.** *If $c_n = D_{\mathrm{sm}} d_n$ for every n, with $D_{\mathrm{sm}} > 0$, then the recurrence, positivity, upper bound and window identity for (c_n) reduce to the same four statements for (d_n) with multiplier B in place of D.*

Height absorption alone does not imply divisibility of a carry state. The identity $DX_a = h_a N - Dv_a$ of Lemma 6.2 is what supplies it for the actual orbit, and Theorem 6.3 is the instance in which the note uses it. The formal consumer takes the common-factor form as a hypothesis.

Proposition 10.2 (conditional extinction of bounded carries)**.** *Let (b_n) and (m_n) be a radix word and a forcing word, let $G : \mathbb{N}_{>0} \times \mathbb{N} \to \mathbb{N}$, and assume cofinal local-window escape against G. Fix $B > 0$ coprime to 30. There is no integral sequence (d_n) satisfying simultaneously $d_{n+1} = b_n d_n - B m_n$, $d_n > 0$ and $|d_n| \leq G(B, n)$ for every $n \geq 0$.*

Proof. Choose one escaping window (ℓ, h). The window identity gives $d_{\ell+h} \equiv -BF_{\ell,h}$ modulo $|W_{\ell,h}|$. The endpoint state is positive and at most $G(B, \ell + h)$, whereas the canonical positive residue of the right-hand side exceeds that bound, so Proposition 7.1 applies. □

Coprimality with 30 is used by the escape hypothesis to select a window; once a window is fixed, the finite contradiction does not use it. The formalisation carries the edge cases: a zero window base is excluded, a zero residue is represented by the full modulus, and positivity prevents the endpoint carry from vanishing. The packaged absorbed form takes a nonzero smooth factor, the exact factorisation $c_n = D_{\mathrm{sm}} d_n$, the absorbed recurrence for (c_n) and the positive short bound for (d_n), and derives the same contradiction in one statement.

10.3 Worked finite examples

The ten smallest values of the running least common multiple at $\{2,3,5\}$ are tabulated in Section 2. They illustrate both parts of Proposition 2.2: the value is constant on $\{5,6,7\}$ and on $\{9,10\}$, and each change multiplies by a single prime, by 2 at $x = 2,4,8$, by 3 at $x = 3,9$ and by 5 at $x = 5$. Also $\mathrm{L}(10) = 8 \cdot 9 \cdot 5 = 360$ is the least common multiple of the smooth numbers $1,2,3,4,5,6,8,9,10$.

For Proposition 2.4 take $(p,q,r) = (2,3,5)$ and the box $\mathcal{B}(1,1,1)$, whose eight points carry the smooth values $1,2,3,5,6,10,15,30$ and the heights

$p^i q^j r^k$	1	2	3	5	6	10	15	30
H	1	2	6	60	60	360	360	10800

Six heights occur, two of them twice: the points 5 and 6 share the height 60, and 10 and 15 share the height 360. The identity reads

$$1 + \tfrac{1}{2} + \tfrac{1}{6} + \tfrac{1}{60} + \tfrac{1}{60} + \tfrac{1}{360} + \tfrac{1}{360} + \tfrac{1}{10800} = 1 + \tfrac{1}{2} + \tfrac{1}{6} + \tfrac{2}{60} + \tfrac{2}{360} + \tfrac{1}{10800} = \tfrac{18421}{10800},$$

and the two coefficients 2 carry the whole content of the regrouping on this box. Where the heights are pairwise distinct the identity is a relabelling.

Multiplying block radices along a run of blocks gives the product of the jump-word letters over that run: for instance $b_1 b_2 = 60$ is the product of the four multipliers at $3,4,5,8$. Block 4 is the only one among the first six carrying an internal power in both channels, and block 5 carries neither, so its radix falls back to the terminal factor alone.

10.4 The finite-scan histogram

The scan of Section 8 covers every $B \le 5000$ coprime to 30 and every start $100 \le \ell \le 3000$, with search depth permitted to 24. It tested 3,869,934 pairs, found an escaping window in every case, and the distribution of first successful lengths was

h	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
#	1	104	812	5437	51409	237423	735450	1431226	1132756	236752	34910	3076	521	49	8

The first case at the maximal observed length 18 is $B = 917$ at start $\ell = 2980$, with endpoint jump index 6179, window base 18139852800000000, forcing 13196471407660025821045, residue 76322101735 and bound 3896420420. An earlier run of the same checker over every $B \le 1000$ coprime to 30 and every start $100 \le \ell \le 500$ tested 106,666 pairs with maximal first successful length 14, whose first case is $B = 359$ at start 291, endpoint jump index 627, base 5038848000000, forcing 25864575212865807, residue 213175287 and bound 15932659.

By Corollary 7.4 these histograms describe a bounded region. The observed concentration at lengths 10 to 12 sits just above the necessary depth $(\log_2 K(B,\ell{+}h) - \log_2 15)/3$ supplied by Proposition 7.3, which is what a reader should expect if the residues behave like generic ones inside this range.

10.5 Source inventory

Each entry names an informal statement and the Lean declaration that carries it. Each link is pinned to its own immutable revision; the exact-reference inventory records the revisions and their comparison with the current public source commit.

informal statement	linked Lean source
smooth lattice value	smooth3 val
pure-power height	three prime height
lattice kernel	three prime kernel q
smooth prefix index set	smooth prefix exponents
running least common multiple	smooth prefix lcm
prefix value divides the height	smooth3 val dvd three prime height of mem
divisibility into the height	smooth prefix lcm dvd three prime height
first pure-power membership	pure first mem smooth prefix exponents
second pure-power membership	pure second mem smooth prefix exponents
third pure-power membership	pure third mem smooth prefix exponents
running-lcm identity	smooth prefix lcm eq three prime height
cell relation	same three prime log cell
cell constancy of the height	three prime height eq of same log cell
cell constancy of the running value	smooth prefix lcm eq of same log cell
cell constancy of the kernel	three prime kernel q eq of same log cell
positive power sets	positive prime powers
channel cardinality	positive prime powers card
exclusion of the origin	one not mem positive prime powers
channel disjointness	positive prime powers disjoint
positive jump channels	three prime positive jump set
positive jump count	three prime positive jump set card
jump set with the origin	three prime jump set with origin
jump count with the origin	three prime jump set with origin card
first height step	three prime height first log step
second height step	three prime height second log step
third height step	three prime height third log step

informal statement	linked Lean source
first coordinate step	smooth prefix lcm first log step
second coordinate step	smooth prefix lcm second log step
third coordinate step	smooth prefix lcm third log step
exponent box	smooth exponent box
point height	smooth point height
height fibre	smooth height fiber
fibre sum	smooth height fiber kernel sum
height-fibre normal form	finite smooth kernel sum grouped by height
cubic majorant	three prime height le cube
origin value	kernel 235 origin
value at two	kernel 235 two
value at three	kernel 235 three
value at six	kernel 235 six
non-separation witness	kernel 235 not rank one
variable-base tail step	tail state step
expanded tail step	tail state step eq
smooth exponent shell	smooth exponent shell
short-interval uniqueness	exponent unique in short interval
first-coordinate projection	smooth exponent shell card le drop first
third-coordinate projection	smooth exponent shell card le drop third
sorted quadratic estimate	sorted pair quadratic
quadratic shell bound	smooth exponent shell card quadratic
dyadic internal power	dyadic internal power
internal-power uniqueness	dyadic internal power exponent unique
dyadic block base	dyadic block base235
exact dyadic radix alphabet	dyadic block base235 cases
bounded-radix consequence	dyadic block base235 mem interval
rank-two certificate	kernel 235 minor eq neg one fifteen
no integer rotation orbit	no integer orbit logb of prime
height factorisation	three prime height factorisation
kernel factorisation	three prime kernel q factorisation
two-prime outer product	two prime kernel q eq outer product
two-prime vanishing minors	two prime kernel q minor two eq zero

informal statement	linked Lean source
uniform minors, general form	exists uniform nonsingular three prime kernel minor
uniform minors for primes	exists uniform nonsingular three prime kernel minor of prime
no finite separation	not finite separable three prime kernel
second-order minor	kernel 235 minor2 eq neg one fifteen
third-order minor	kernel 235 minor3 eq one over 81000
misleading proportional row	kernel 235 row three eq smul row zero
failure of that proportionality	kernel 235 row three ne smul row zero at four
ordered block digit	dyadic ordered block digit235
normalised state step	dyadic normalized tail state r235 succ
shell summability	summable dyadic shell mass r235
infinite tail recurrence	dyadic normalized shell tsum tail r235 succ
integer-or-cofinally-far dichotomy	dyadic shell tsum tail integer or cofinal far
boundary divisibility	smooth height mul prime dvd boundary height
half-height identity	two mul height normalizer235
window clearing identity	height normalizer235 mul window mass eq int
all-scale rationality lattice	qsmul normalized tail state eq int of value eq rat
normalised-state collision	exists normalized tail state collision of value eq rat
least positive residue	least positive residue
positive representative range	least positive residue pos le
representative congruence	least positive residue mod eq
escape condition	residue escapes window
finite natural-state contradiction	no bounded positive state of residue escape
contrapositive form	residue le bound of bounded positive state
integer least-positive-residue obstruction	no bounded positive int state of least positive residue
exact residue classifier	least positive residue eq nat abs of pos le mod eq
window base	window base
window forcing	window forcing
affine window identity	affine recurrence window
scaled forcing identity	window forcing const mul

informal statement	linked Lean source
integral carry window	integral carry window
endpoint residue identification	least positive residue window forcing eq carry
absorbed smooth factor	smooth3 val dvd three prime height of le
common-factor cancellation	integral carry cancel common factor
reduced bound transfer	reduced carry pos le of common factor
reduced window transfer	reduced integral carry window
cofinal window hypothesis	cofinal local window escape
reduced-carry extinction	no positive reduced carry of cofinal local window escape
absorbed-carry extinction	no positive absorbed carry of cofinal local window escape
real window identity	true normalized state window
escape from irrationality, general bound	cofinal local window escape of irrational of beaten
escape from irrationality, quadratic family	cofinal local window escape of irrational of quadratic
escape from irrationality, actual bound	cofinal local window escape of irrational
producer equivalence, shifted tail	actual cofinal local window escape iff
producer equivalence, series value	actual cofinal local window escape iff irrational value
rationality-to-carry bridge	exists reduced carry of value eq rat
the actual producer	actual cofinal local window escape
irrationality from the producer	irrational value of cofinal local window escape
bounded-radix alternative	bounded radix zero or cofinal far
rational value from an integral scaled tail	rational of scaled tail integer
channel potential classifier	channel block null iff channel potential
one-index lift extinction	no carry lift of error bound below two pow
uniform lift extinction	no bounded carry lift of block null two anchors
first-block sum	binary lift two anchors force first block sum one

informal statement	linked Lean source
four-state obstruction	no unit accurate lift with two anchors and first two block null
residue-coboundary form	carry eq residue digit add coboundary
carry interval	carry residue mem interval
digit interval	residue digit mem interval
symbolic realisation	cartier realisation
finite realised span	finite realised span of factorisation
Cantor-state confinement	state mem ioo
state gap bound	state gap lower bound
small-form criterion	irrational of small nonzero forms
conditional assembly	irrational of clearing and small gaps
staircase index engine	staircase indices

11 Complete result-family map

This section places every registered family for this problem in the shared 70-family reader order. The five display bands control exposition only; the separate promotion state currently covers 6 families and is reported but does not hide or strengthen any family. Mathematical statements and evidence modes come from the public claim registry. Across all eight problems the public result-atom catalog contains 681 exact packet coordinates; this problem contributes 65. The recovered source catalog contains 682 rows; 1 row(s) belong to families absent from the current claim registry and remain disclosed as detached source rows. Catalog rows expose bounded statement excerpts plus full-source digests, not a claim that every complete packet statement is reproduced here.

11.1 Three prime lcm cells

Reader position. 6 of 70; display band: front door. Formal editorial disposition: merge. These are separate classifications.

Reader entry. Exact running-LCM product, logarithmic-cell constancy, coordinate jump ratios, and jump count.

Exact running-LCM product, logarithmic-cell constancy, coordinate jump ratios, and jump count.

Authority and reach. Lean kernel; Comparator-selected; locally proved result; novelty unassessed.

Exact boundary. Cell structure alone does not prove irrationality.

Result-atom population. 5 of 681 public coordinates. The public atom catalog groups them under this family.

Comparator assurance. exact selected interface; 1 executable interface(s), 1 repository-registered selected result interface(s).

Publication placement. This family is also admitted to the short note.

11.2 Two prime transcendence

Reader position. 17 of 70; display band: major result. Formal editorial disposition: hold. These are separate classifications.

Reader entry. Both two-prime running-lcm series are proved transcendental by an authored deduction from an external theorem.

Both two-prime running-lcm series are proved transcendental by an authored deduction from an external theorem.

Authority and reach. paper argument plus cited theorem; paper plus external theorem.

Exact boundary. Comparator cannot certify the external analytic input.

Result-atom population. 1 of 681 public coordinates. The public atom catalog groups them under this family.

Comparator assurance. not applicable to comparator; 0 executable interface(s), 0 repository-registered selected result interface(s).

Publication placement. This family is also admitted to the short note.

11.3 Weighted phase carry observer

Reader position. 24 of 70; display band: mechanism. Formal editorial disposition: hold. These are separate classifications.

Reader entry. An exact weighted-phase carry recurrence splits into a finite residue digit and an uncontrolled integral quotient coboundary; an explicit function-faithful finite-dimensional observer then forces finite realised span.

An exact weighted-phase carry recurrence splits into a finite residue digit and an uncontrolled integral quotient coboundary; an explicit function-faithful finite-dimensional observer then forces finite realised span.

Authority and reach. Lean kernel; locally proved result; novelty unassessed.

Exact boundary. The recurrence supplies a finite residue coordinate but leaves an uncontrolled integral quotient coboundary; it proves neither a finite-state quotient nor a literal infinite realised span. Finite realised span requires an explicit factorisation through a finite-dimensional function-faithful observer, and scalar evaluation alone is insufficient. No irrationality conclusion follows from this observer alone. The actual rationality-to-positive-reduced-carry bridge is established separately; cofinal local-window escape remains open.

Result-atom population. 7 of 681 public coordinates. The public atom catalog groups them under this family.

Comparator assurance. exact selected interface; 1 executable interface(s), 0 repository-registered selected result interface(s).

Publication placement. This family remains in the complete long record and is not a short-note headline.

- `ErdosProblems.Erdos269.carry_eq_residueDigit_add_coboundary`
- `ErdosProblems.Erdos269.carryResidue_mem_interval`
- `ErdosProblems.Erdos269.residueDigit_mem_interval`
- `ErdosProblems.Erdos269.finite_realisedSpan_of_factorisation`

11.4 Height fibre and shell

Reader position. 28 of 70; display band: mechanism. Formal editorial disposition: merge. These are separate classifications.

Reader entry. Finite height-fibre normal form and a quadratic smooth-shell multiplicity bound.

Finite height-fibre normal form and a quadratic smooth-shell multiplicity bound.

Authority and reach. Lean kernel; locally proved result; novelty unassessed.

Exact boundary. These fibre bounds are counting input, not an escape producer. The separate actual-series divisibility bridge is established.

Result-atom population. 5 of 681 public coordinates. The public atom catalog groups them under this family.

Comparator assurance. represented by selected interface; 0 executable interface(s), 0 repository-registered selected result interface(s).

Publication placement. This family remains in the complete long record and is not a short-note headline.

- `ErdosProblems.Erdos269.finiteSmoothKernelSum_groupedByHeight`
- `ErdosProblems.Erdos269.smoothExponentShell_card_quadratic`

11.5 Conditional carry escape

Reader position. 42 of 70; display band: frontier. Formal editorial disposition: hold. These are separate classifications.

Reader entry. Under the denominator-dependent cofinal local-window residue-escape predicate, no positive reduced carry can satisfy the exact multiplier recurrence together with its short bound. The load-bearing consumer is no_positive_reducedCarry_of_cofinalLocalWindowEscape; an absorbed nonzero common-factor carry reduces exactly to that consumer. CofinalLocalWindowEscape, windowBase, windowForcing, leastPositiveResidue, and the absorbed-carry bridge are subordinate finite-window mechanism evidence.

Under the denominator-dependent cofinal local-window residue-escape predicate, no positive reduced carry can satisfy the exact multiplier recurrence together with its short bound. The load-bearing consumer is no_positive_reducedCarry_of_cofinalLocalWindowEscape; an absorbed nonzero common-factor carry reduces exactly to that consumer. CofinalLocalWindowEscape, windowBase, windowForcing, leastPositiveResidue, and the absorbed-carry bridge are subordinate finite-window mechanism evidence.

Authority and reach. Lean kernel; Comparator-selected; conditional no-go consumer; novelty and significance unassessed.

Exact boundary. The representative assumes b,m : $\mathbb{N} \to \mathbb{N}$, shortBound : $\mathbb{N} \to \mathbb{N} \to \mathbb{N}$, CofinalLocalWindowEscape b m shortBound, and for each positive B coprime to 30 a positive integer-valued d with $d(n+1) = (b\ n : \mathbb{Z})\ d(n) - (B : \mathbb{Z})\ (m\ n : \mathbb{Z})$ and Int.natAbs (d n) $\leq$ shortBound B n; it then gives False. The absorbed-carry bridge additionally assumes smoothFactor $\neq 0$, c n = smoothFactor * d n, and the corresponding exact absorbed recurrence, then cancels that factor to the same reduced consumer. The representative remains a conditional consumer for arbitrary words. For the actual three-prime word, the rationality-to-reduced-carry bridge is established separately; the source-specific cofinal escape producer is not proved. This is not a #269 endpoint or irrationality proof, and no actual-series identification, novelty, priority, significance, or external-review claim is made. It is one conditional window-carry family, distinct from the finite residue, rank-two, and weighted-phase observer families.

Result-atom population. 37 of 681 public coordinates. The public atom catalog groups them under this family.

Comparator assurance. exact selected interface; 1 executable interface(s), 1 repository-registered selected result interface(s).

Publication placement. This family is also admitted to the short note.

- `ErdosProblems.Erdos269.CofinalLocalWindowEscape`

- `ErdosProblems.Erdos269.no_positive_reducedCarry_of_cofinalLocalWindowEscape`
- `ErdosProblems.Erdos269.windowBase`
- `ErdosProblems.Erdos269.windowForcing`
- `ErdosProblems.Erdos269.leastPositiveResidue`
- `ErdosProblems.Erdos269.no_positive_absorbedCarry_of_cofinalLocalWindowEscape`

11.6 Rank two kernel no go

Reader position. 54 of 70; display band: frontier. Formal editorial disposition: hold. These are separate classifications.

Reader entry. The 2,3,5 kernel is not rank one and its smallest displayed minor equals -1/15.

The 2,3,5 kernel is not rank one and its smallest displayed minor equals -1/15.

Authority and reach. Lean kernel; Comparator-selected; no-go result.

Exact boundary. Failure of rank one does not itself imply irrationality.

Result-atom population. 3 of 681 public coordinates. The public atom catalog groups them under this family.

Comparator assurance. exact selected interface; 1 executable interface(s), 1 repository-registered selected result interface(s).

Publication placement. This family is also admitted to the short note.

11.7 Dyadic block alphabet

Reader position. 62 of 70; display band: technical support. Formal editorial disposition: merge. These are separate classifications.

Reader entry. The exact dyadic block alphabet is 2, 6, 10, and 30.

The exact dyadic block alphabet is 2, 6, 10, and 30.

Authority and reach. Lean kernel; locally proved result; novelty unassessed.

Exact boundary. The finite alphabet does not supply the needed carry escape.

Result-atom population. 5 of 681 public coordinates. The public atom catalog groups them under this family.

Comparator assurance. represented by selected interface; 0 executable interface(s), 0 repository-registered selected result interface(s).

Publication placement. This family remains in the complete long record and is not a short-note headline.

- `ErdosProblems.Erdos269.dyadicBlockBase235_cases`

11.8 Three prime finite search

Reader position. 64 of 70; display band: technical support. Formal editorial disposition: hold. These are separate classifications.

Reader entry. A local-window checker searches 106666 denominator and start pairs and records small certificates.

A local-window checker searches 106666 denominator and start pairs and records small certificates.

Authority and reach. external exact computation; finite computation.

Exact boundary. Finite search is not a cofinal statement.

Result-atom population. 2 of 681 public coordinates. The public atom catalog groups them under this family.

Comparator assurance. not applicable to comparator; 0 executable interface(s), 0 repository-registered selected result interface(s).

Publication placement. This family remains in the complete long record and is not a short-note headline.

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